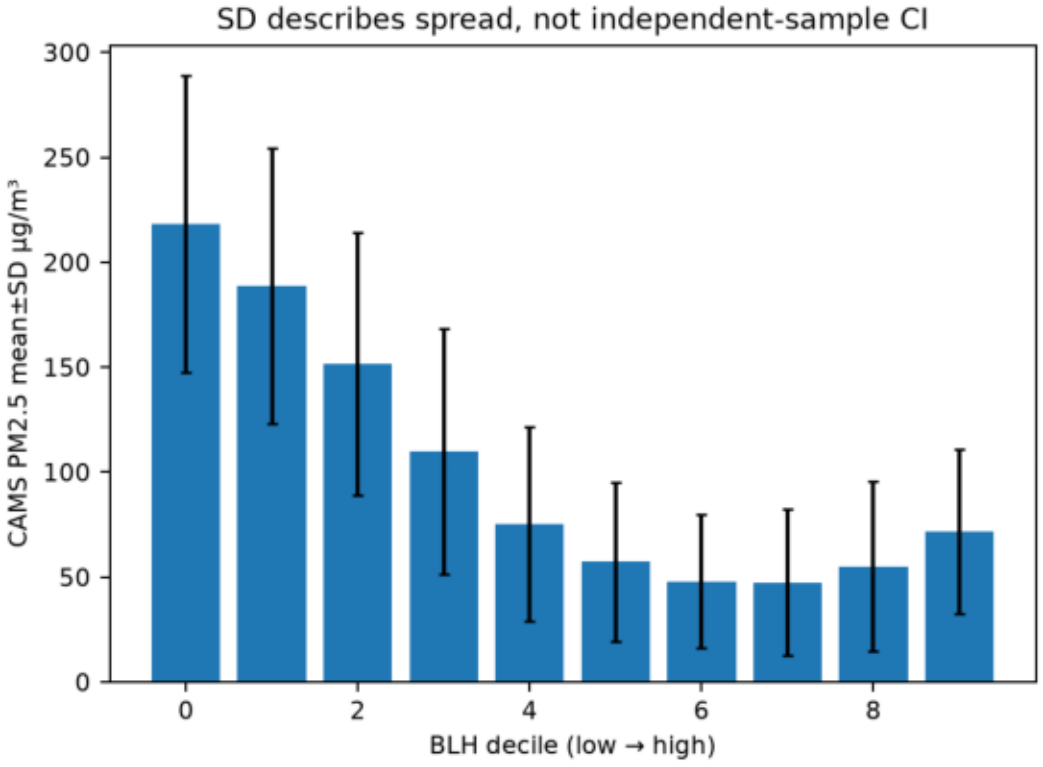
Dhaka air-quality analysis: executed results

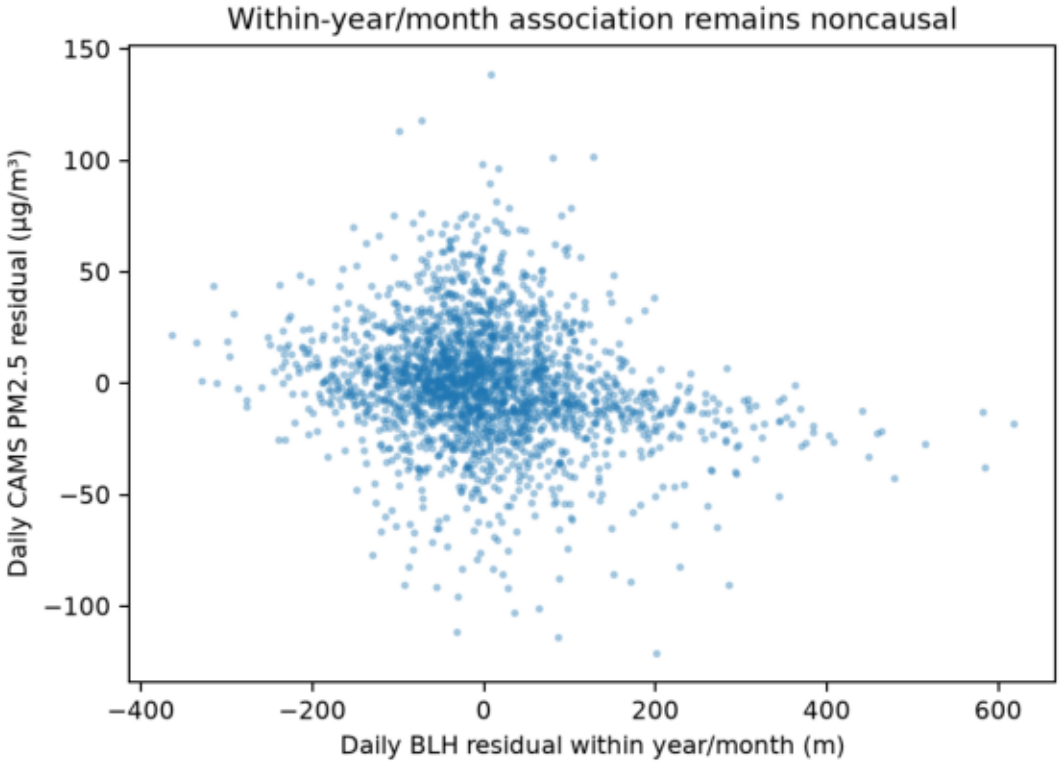
Final-release reproduction + measured reference + conditional prediction

- Observed reference: 2117/2192 days; mean 87.18 µg/m³.
- Winter 172.02 versus June–September 33.53 µg/m³.
- Trend 0.677 µg/m³/year; block interval [-0.166, 1.533] spans zero.
- Conditional regional/monitor comparison: n=1752, r=0.897, bias +16.15, RMSE 34.23 µg/m³.
- 2020–21 conservative prediction: persistence RMSE 29.51; core Ridge 27.74; lag2-weather 27.48.
- Paired weather-minus-core RMSE interval [-0.810, 0.382]: added weather benefit not established.
- Final HyAQMet hourly BLH r=-0.4904, not the printed -0.65; threefold decile contrast reproduces.
- Season-label mismatch and released precipitation accumulation concern remain unresolved.
- Daily averaging/QA and publication latency limit interpretation; no causal, health-outcome or operational claim.

Detailed methods: ANALYSIS_REPORT.md. Data attribution: Barua/Shuvo et al.; EPA/Sarwar et al.; NASA POWER; AirNow mirror creators.

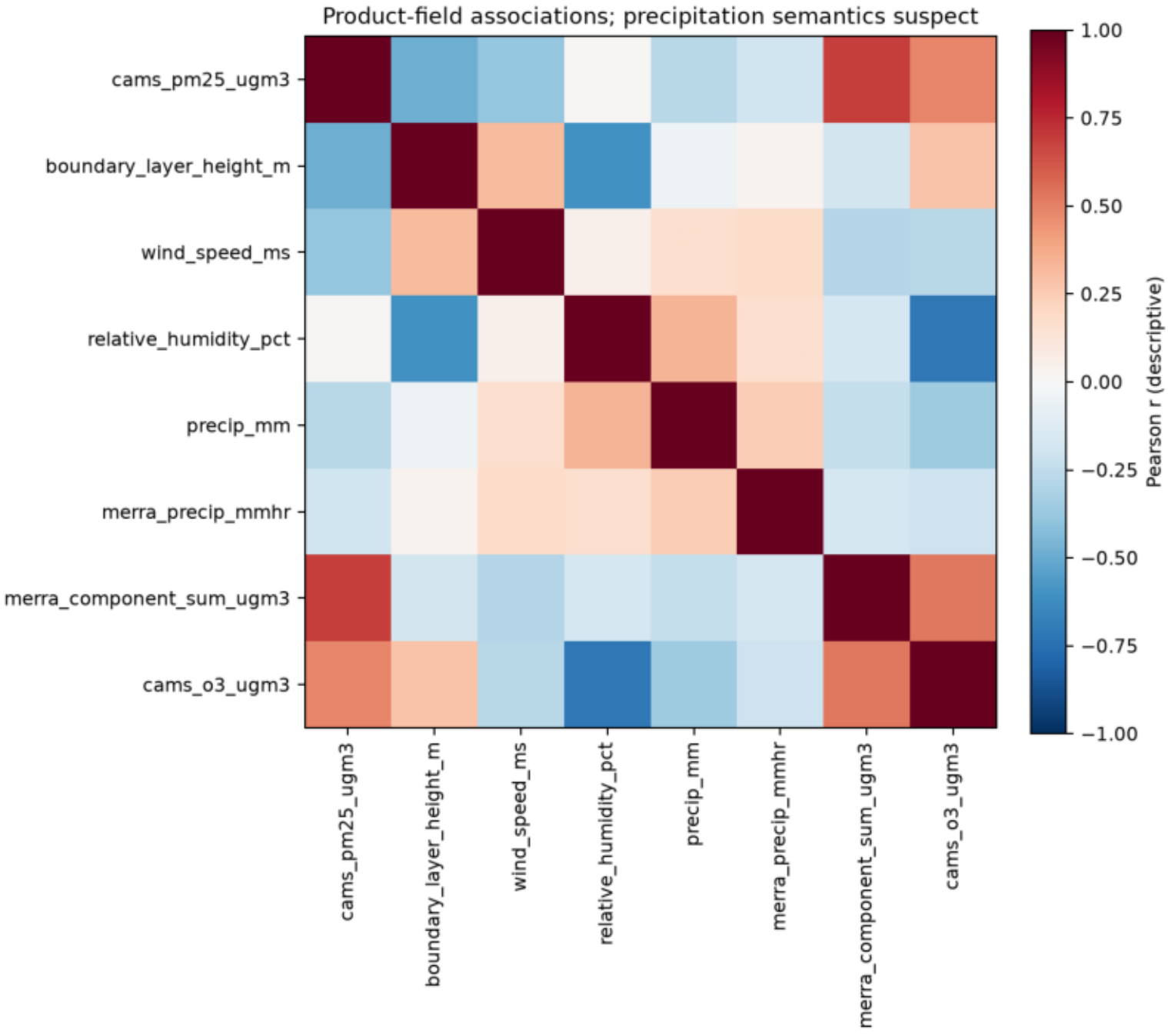
hyaqmet/blh_association.png





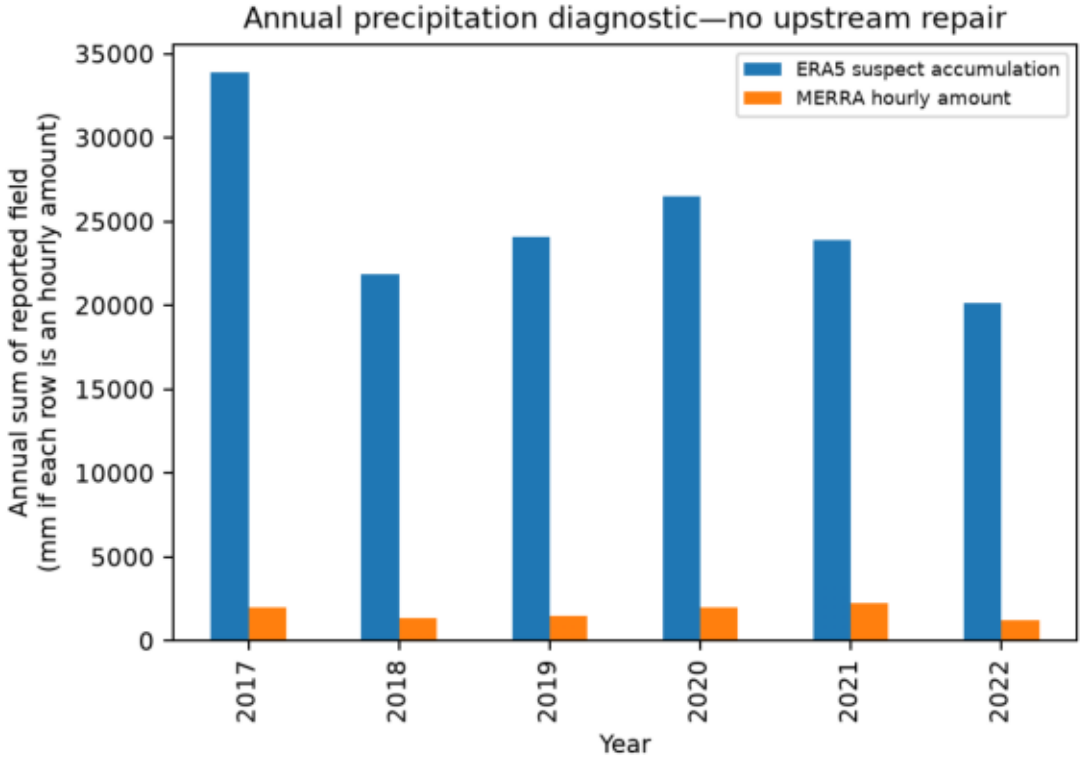
Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

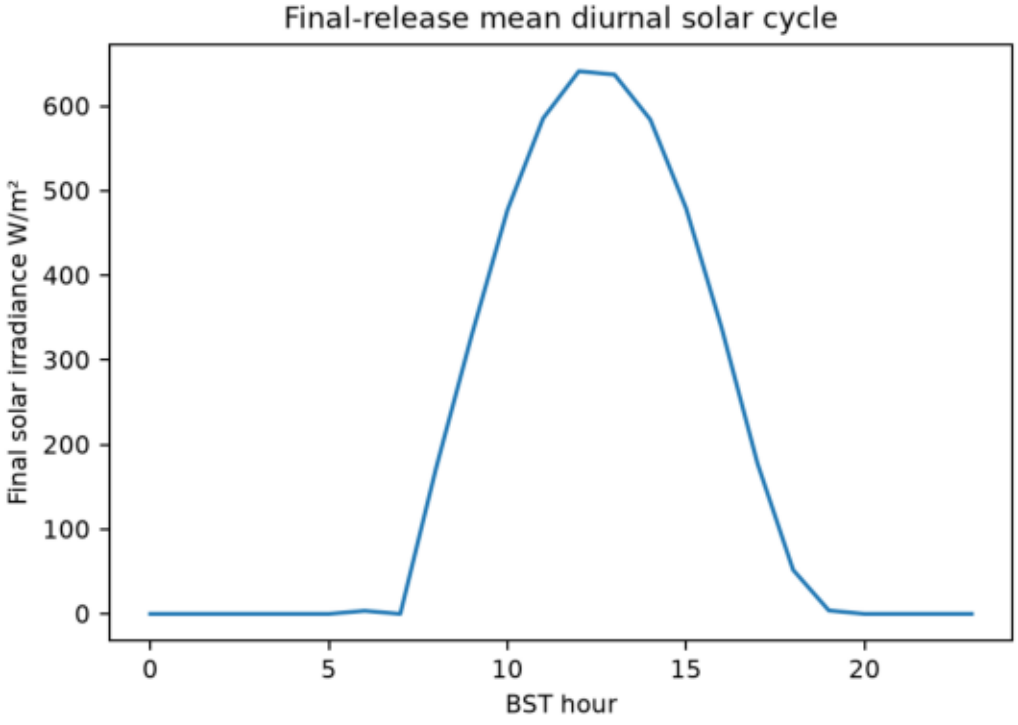
hyaqmet/correlation.png



Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

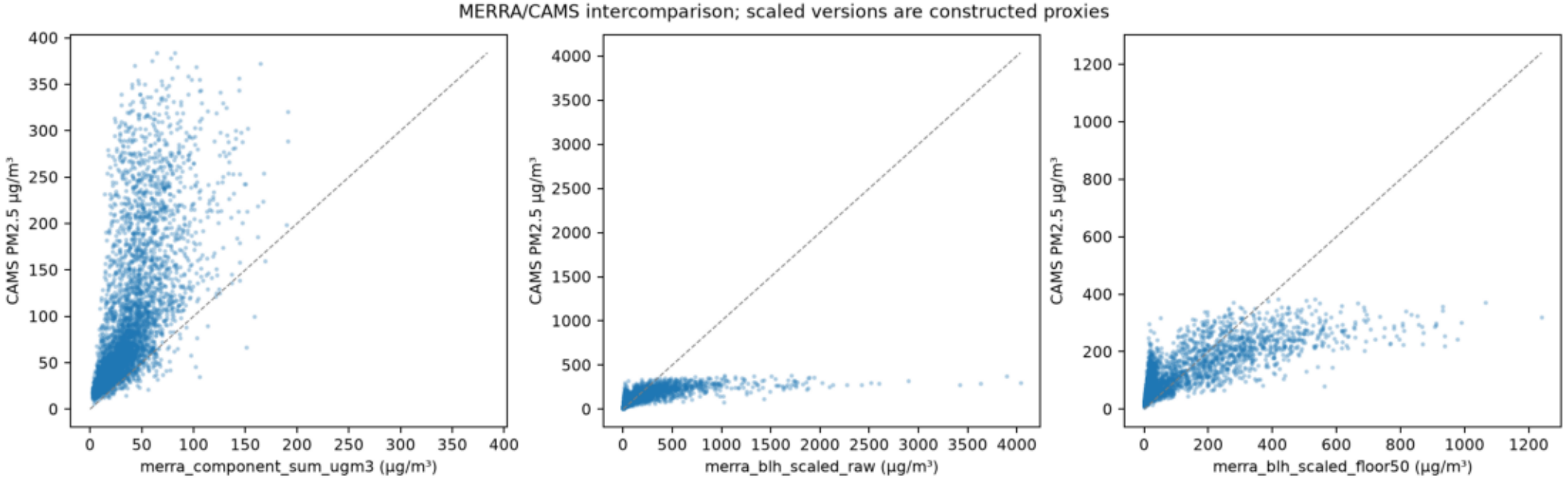
hyaqmet/data_quality.png





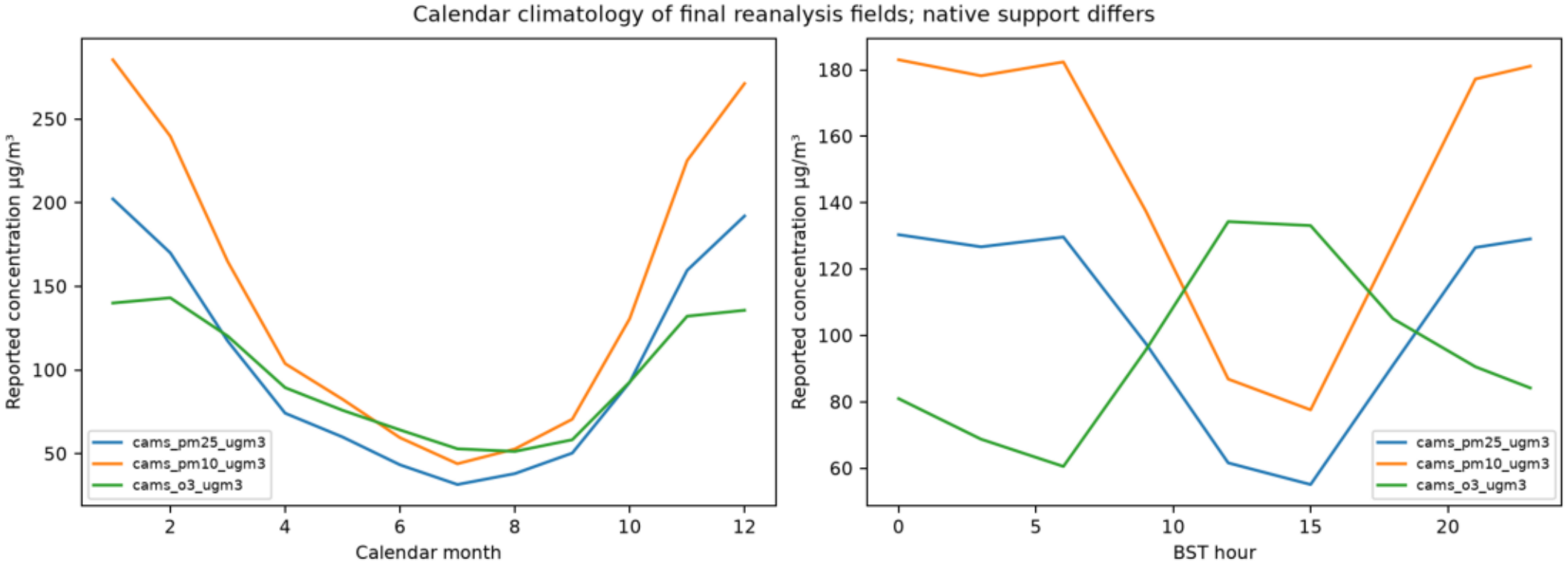
Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

hyaqmet/pm25_cross_source.png



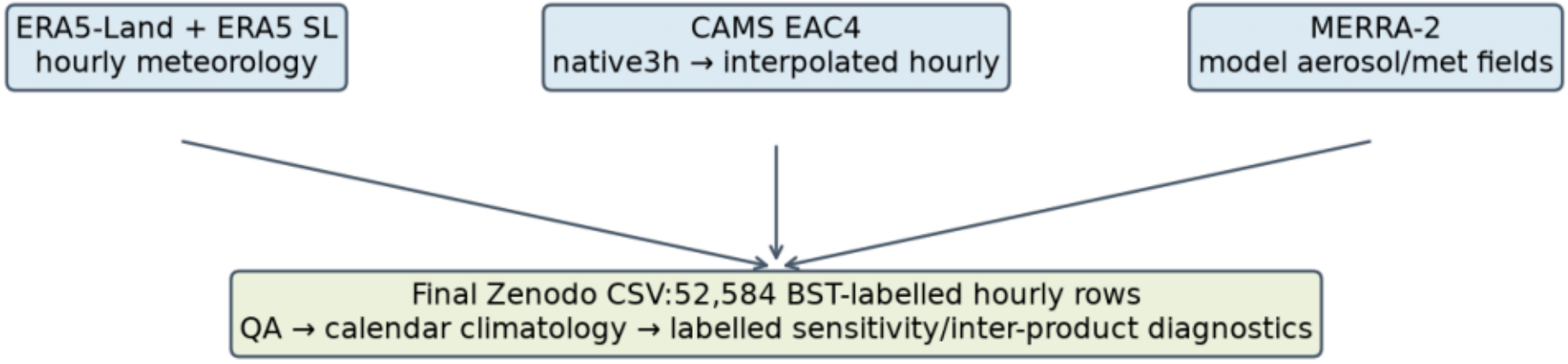
Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

hyaqmet/pollutant_climatology.png



Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

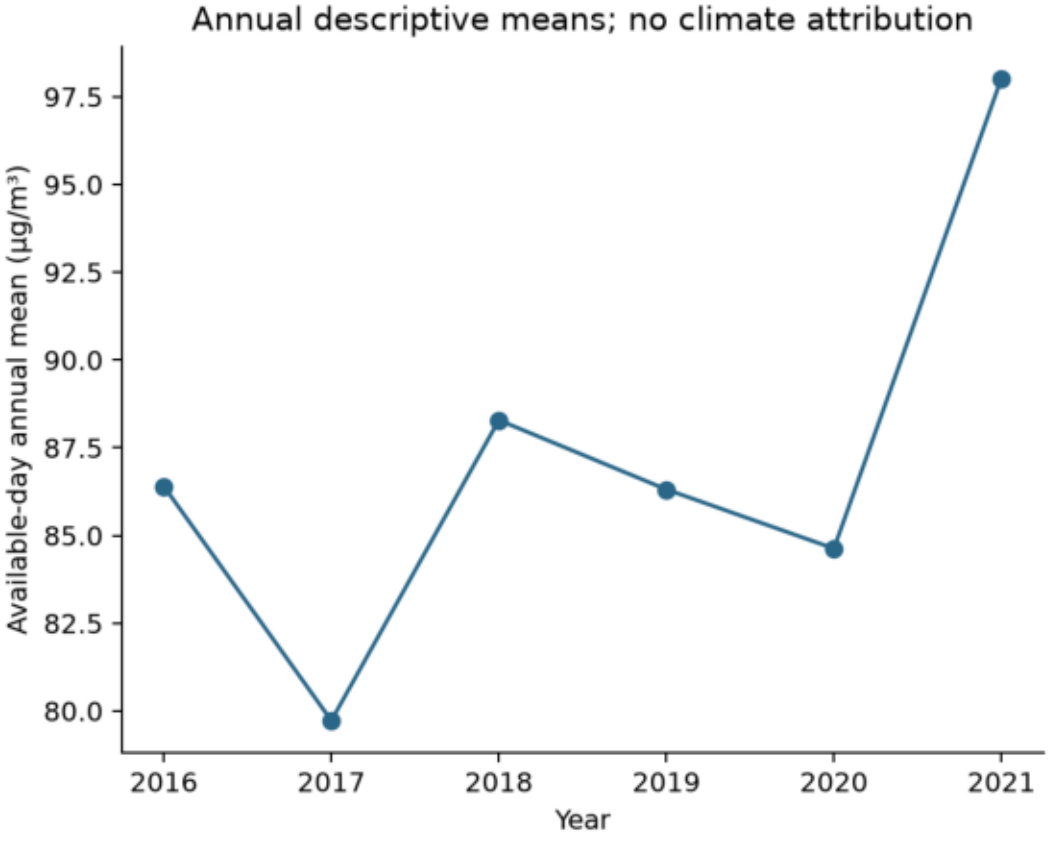
hyaqmet/workflow.png

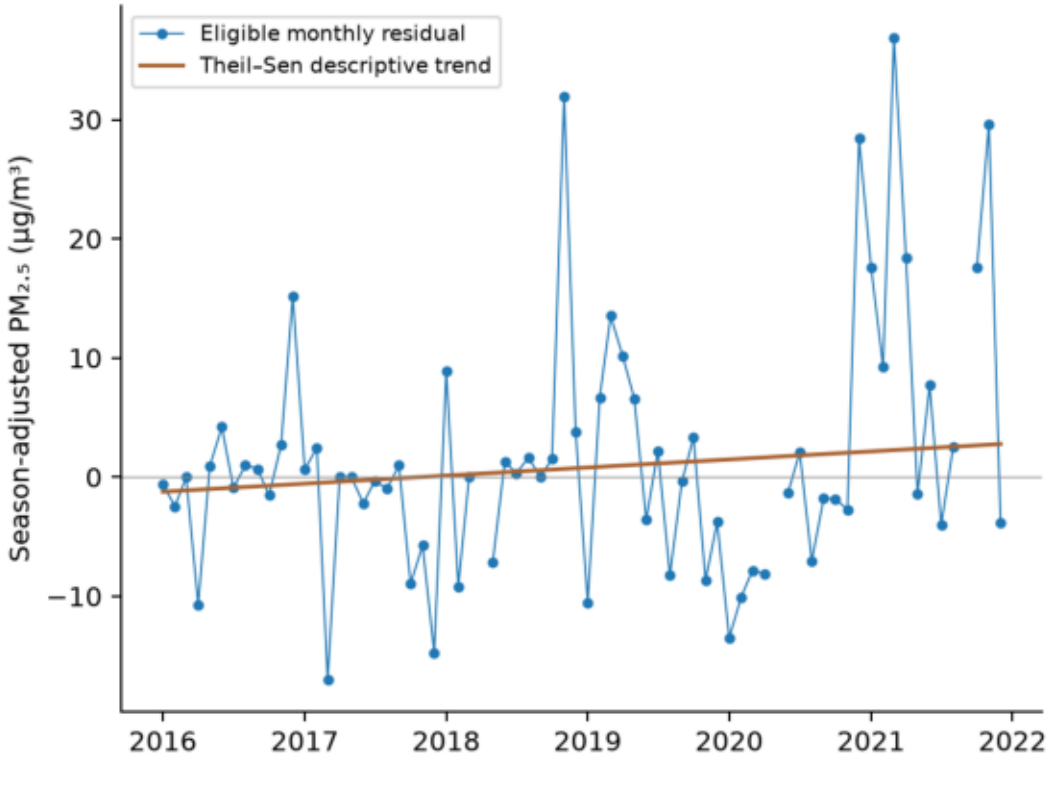


Reanalysis agreement ≠ independent station validation; source defects remain flagged

Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

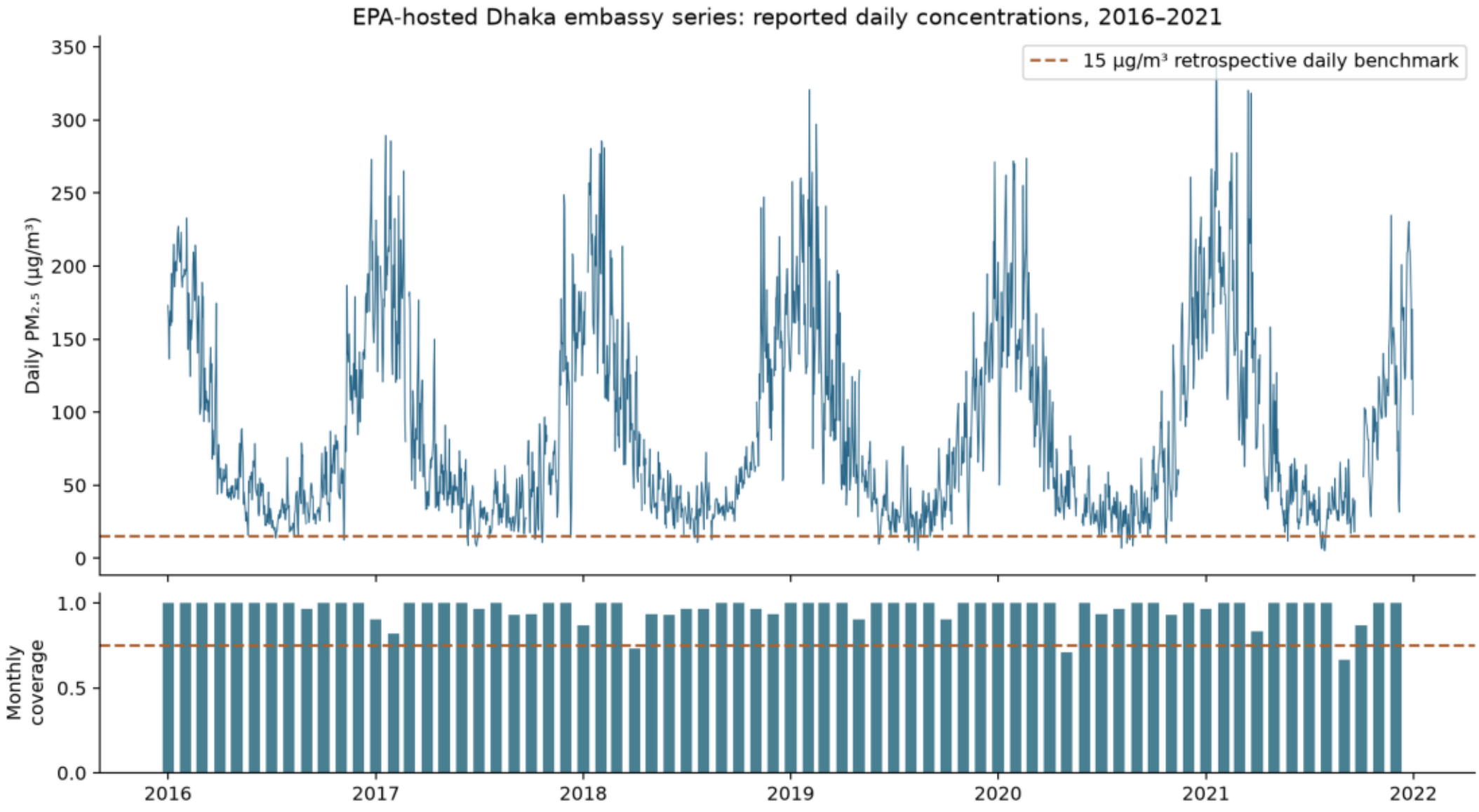
observed/observed_annual_and_adjusted_trend.png





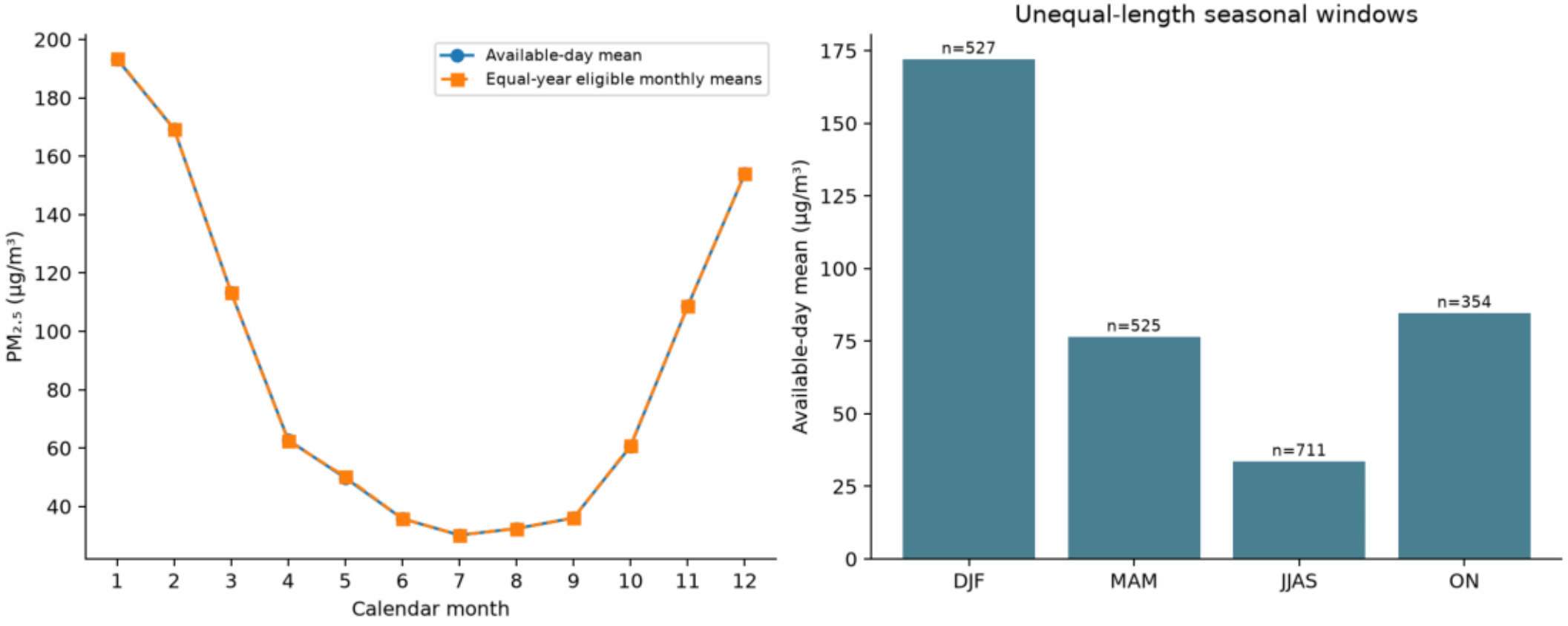
Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

observed/observed_daily_and_coverage.png



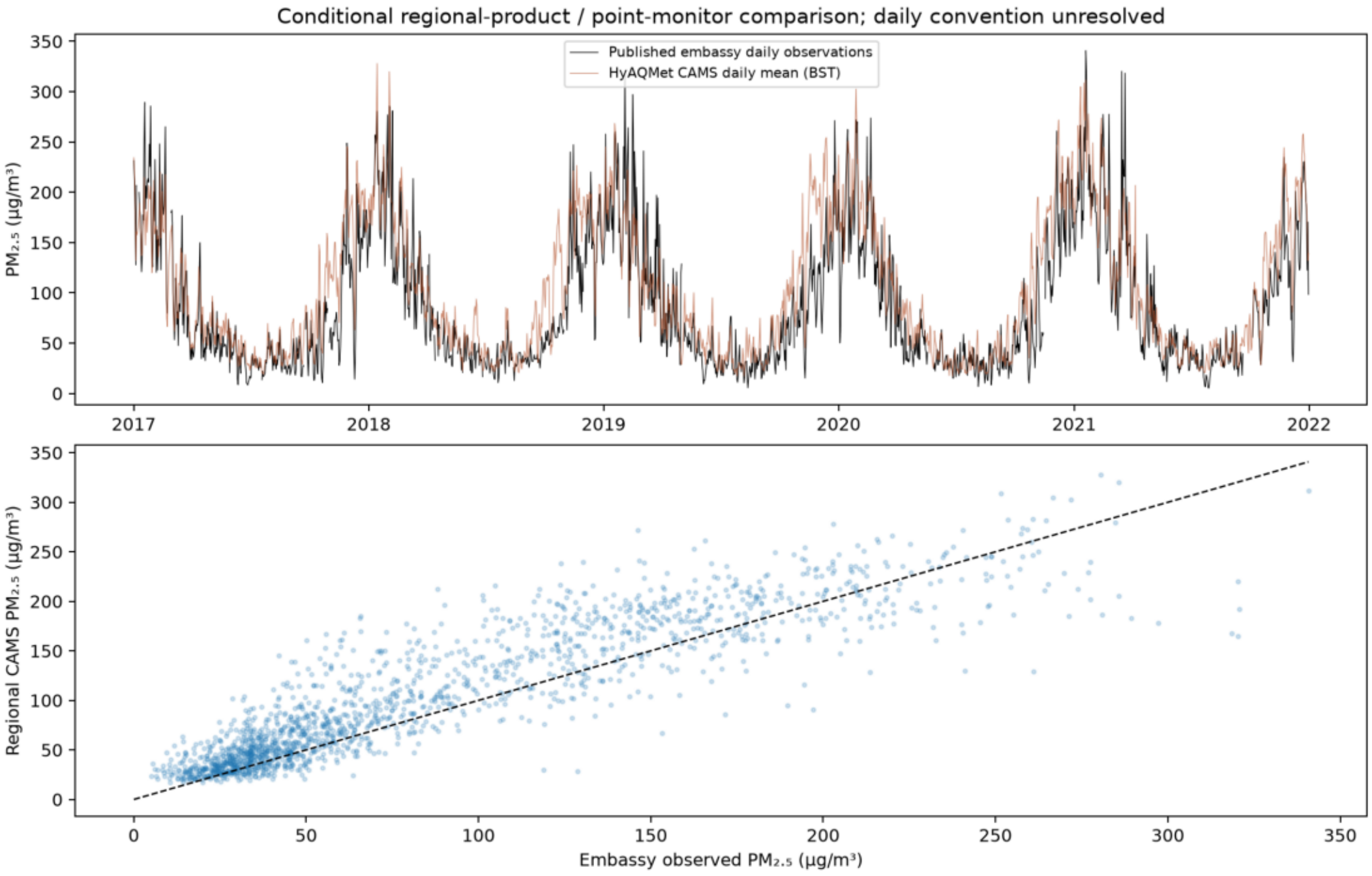
Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

observed/observed_monthly_and_seasonal.png



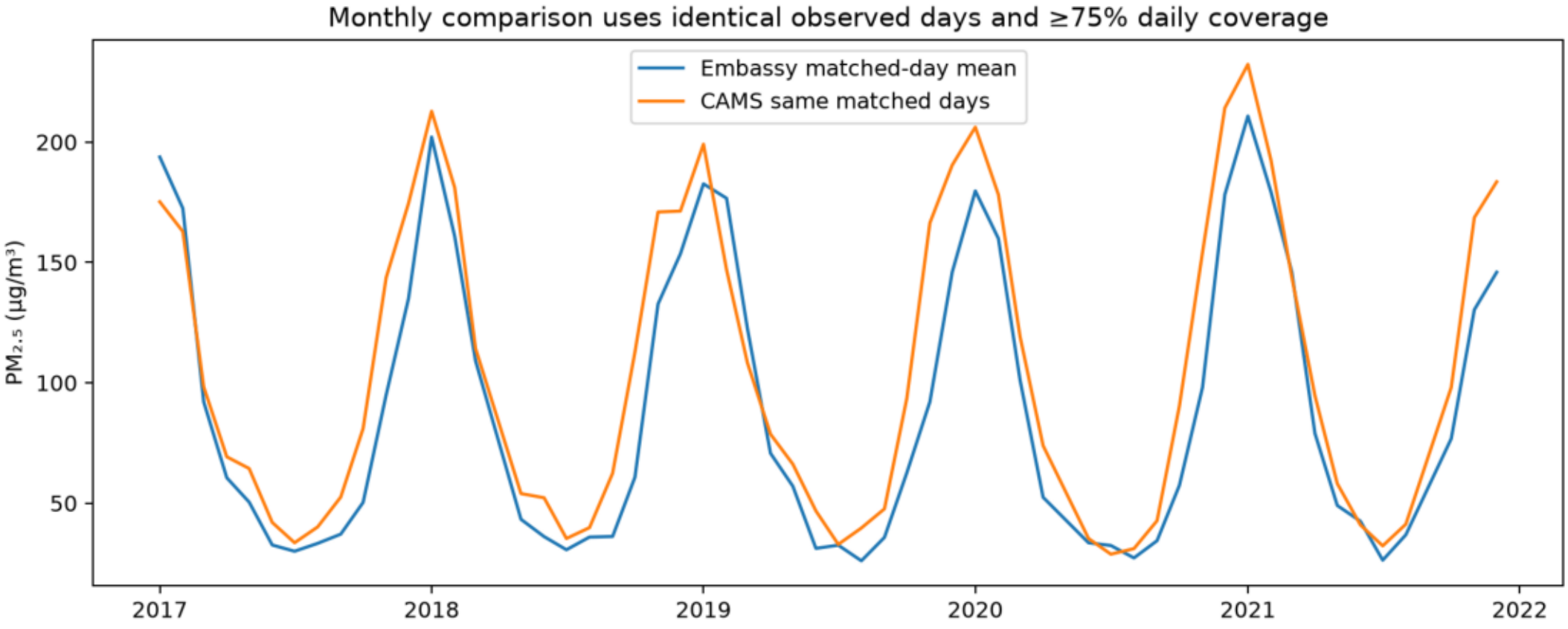
Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

reference/independent_reference_comparison.png



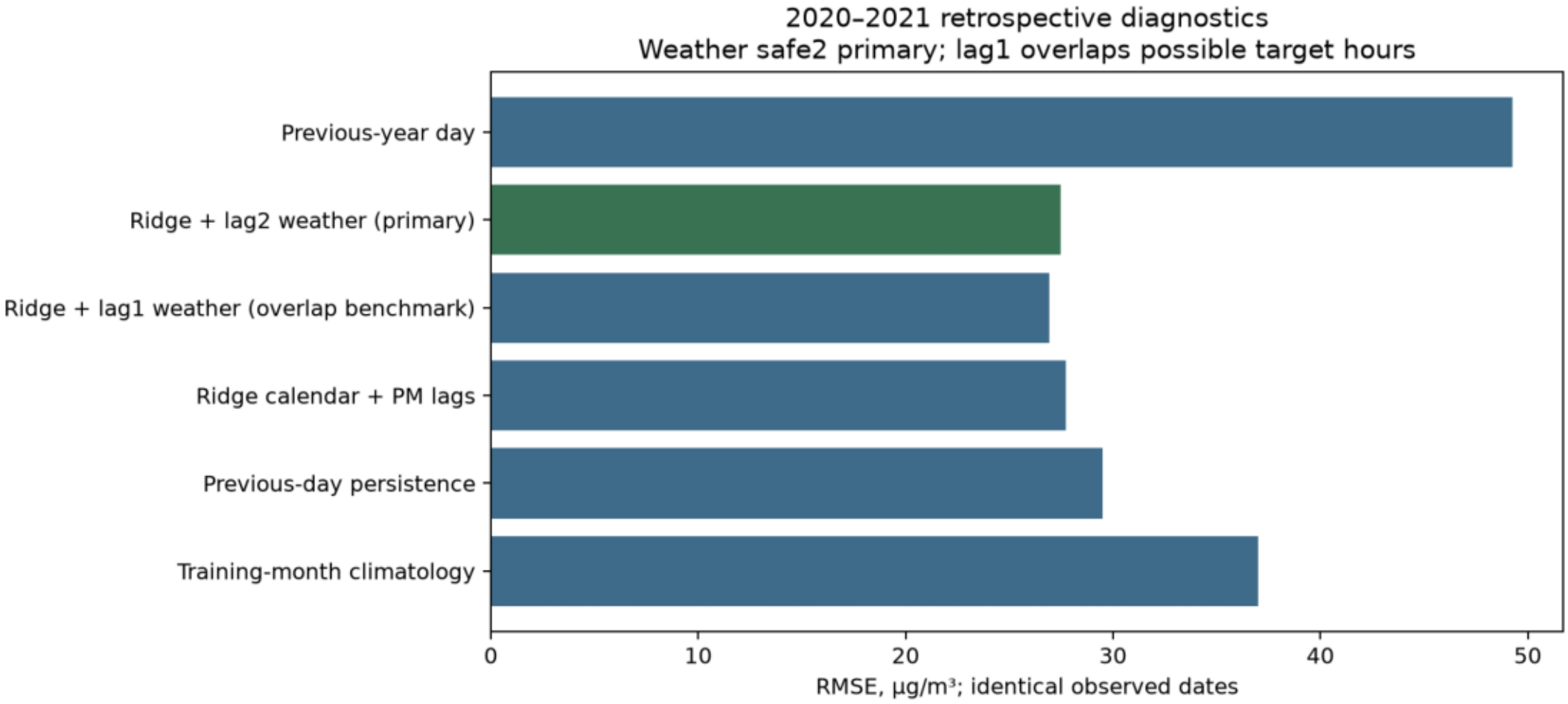
Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

reference/matched_monthly_comparison.png



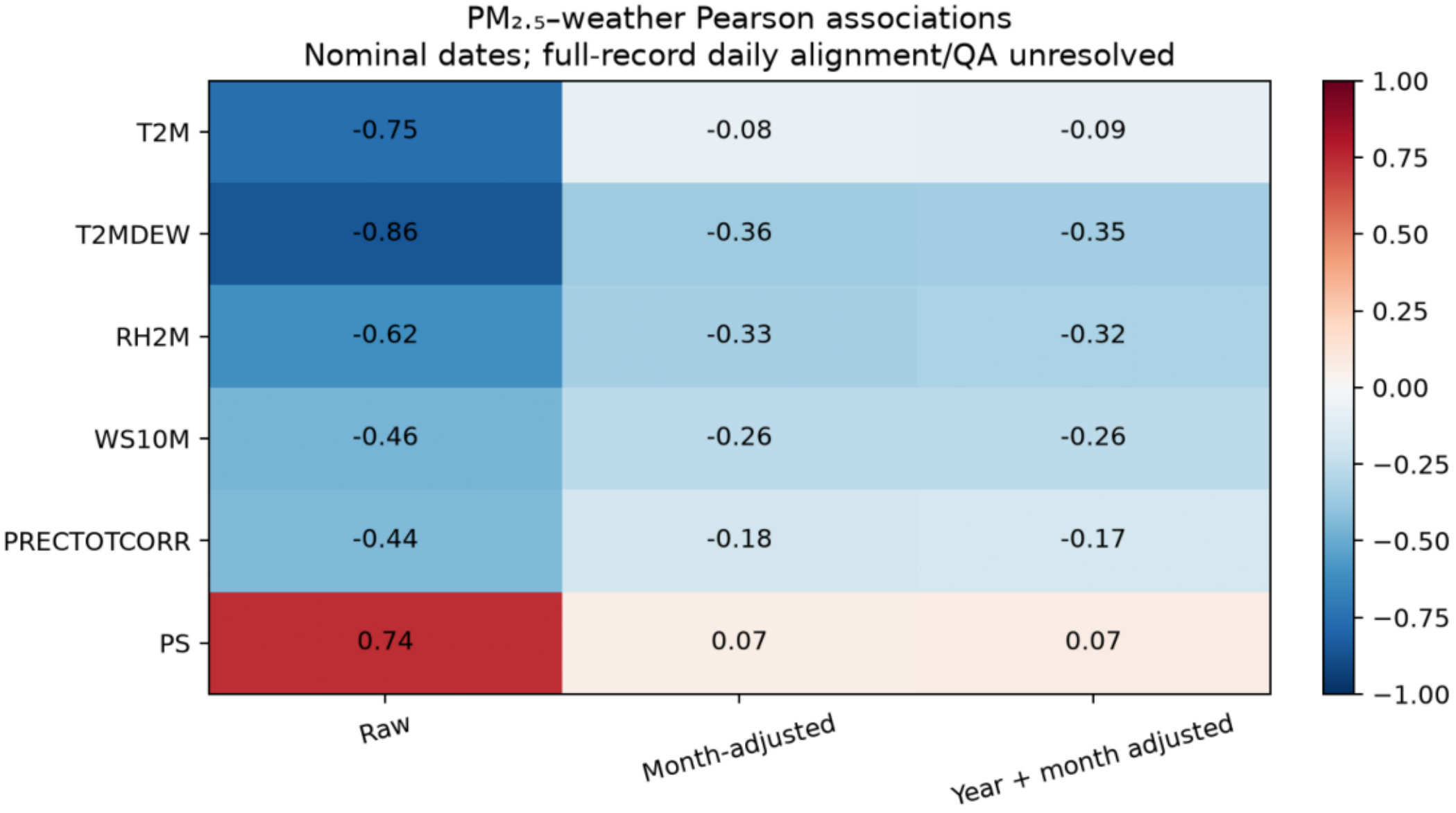
Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

weather/daily_model_comparison.png



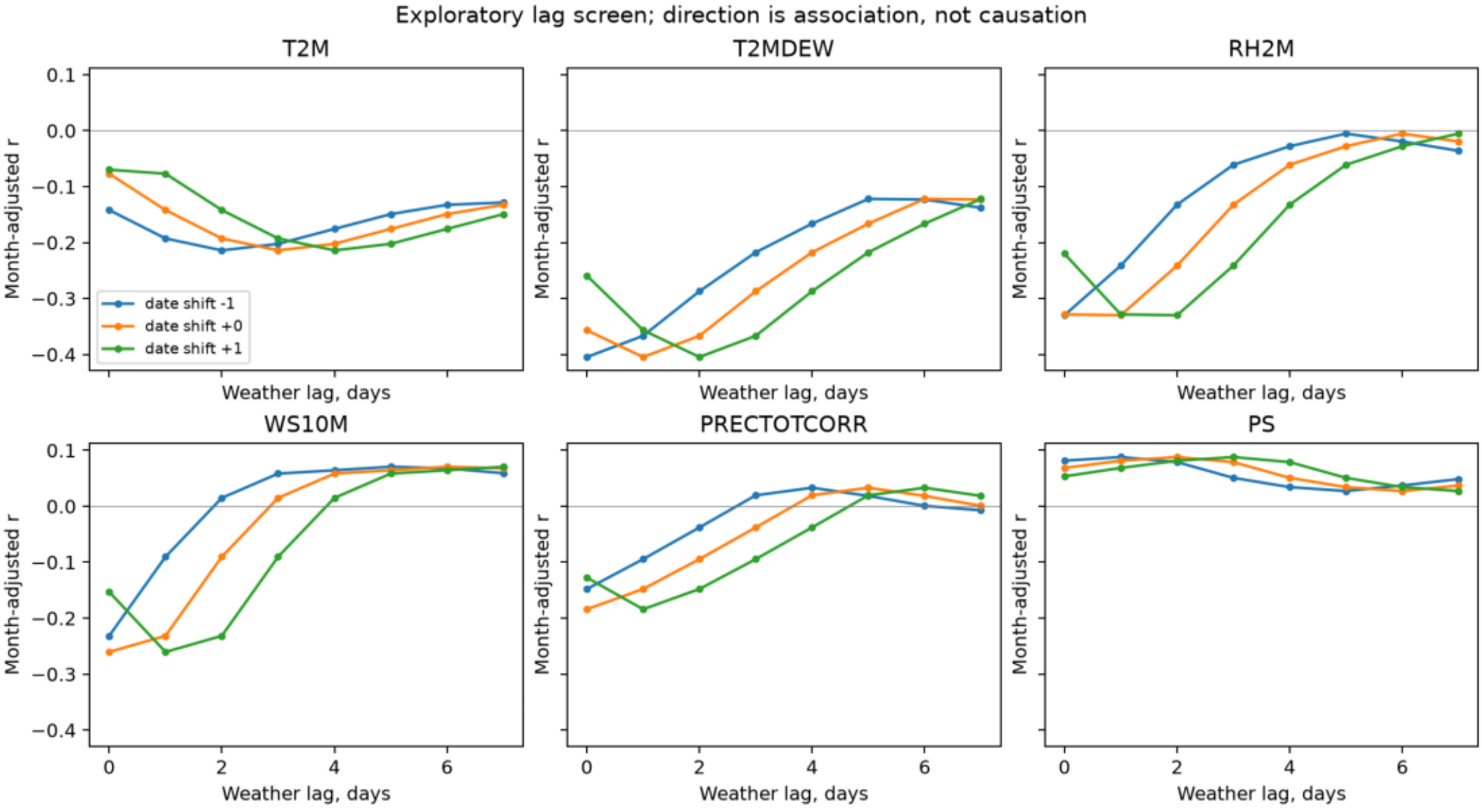
Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

weather/weather_correlation_heatmap.png



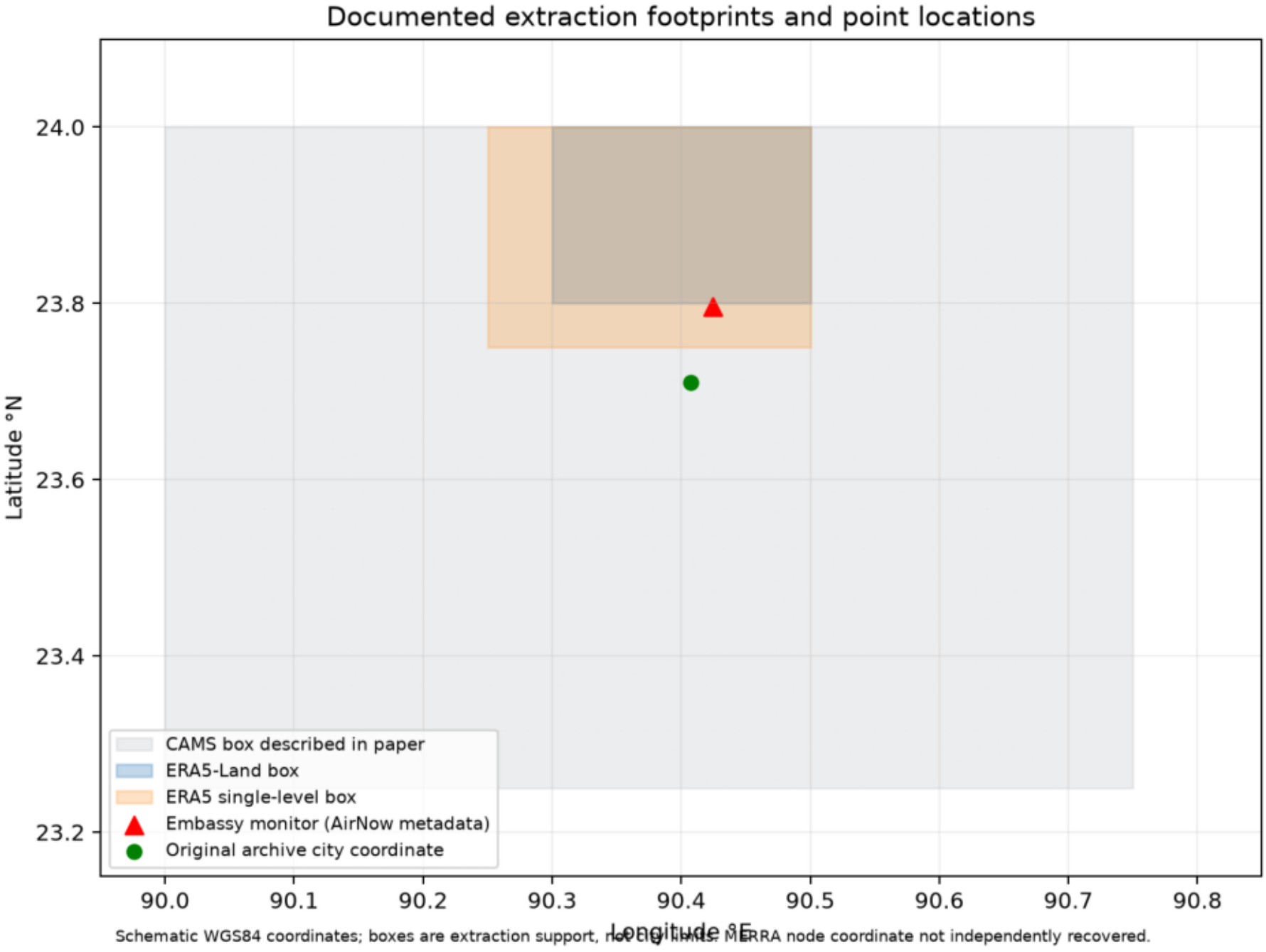
Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

weather/weather_lag_sensitivity.png



Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.

source_support_schematic.png



Read the accompanying report and per-track methods before interpretation; figures retain their source/conditional-analysis scope.